

$$a) \int \frac{\sin x}{1 + \sin x + \cos x} dx = ?$$

$$\tan \frac{x}{2} = t \Rightarrow dx = \frac{2dt}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, \sin x = \frac{2t}{1+t^2}$$

$$\frac{2t}{(1+t)(1+t^2)} = \frac{A}{1+t} + \frac{Bt+C}{1+t^2}$$

$$A = -1, B = C = 1$$

$$I = -\int \frac{1}{1+t} dt + \int \frac{t}{1+t^2} dt + \int \frac{1}{1+t^2} dt$$

$$= -\ln|1+t| + \frac{1}{2} \ln(1+t^2) + \tan^{-1} t + C_1$$

$$= -\ln\left|1 + \tan \frac{x}{2}\right| + \frac{1}{2} \ln\left(1 + \tan^2 \frac{x}{2}\right) + \frac{x}{2} + C$$

Use the substitution $u = e^t$ and then the partial fraction method to find the antiderivative

$$\int \frac{e^{4t} + 2e^{2t} - e^t}{e^{2t} + 1} dt$$

Using the substitution

$$u = e^t, \quad du = e^t dt \quad \text{or} \quad t = \ln u, \quad dt = \frac{1}{u} du = e^{-t} du$$

the integral becomes

$$\int \frac{e^{4t} + 2e^{2t} - e^t}{e^{2t} + 1} dt = \int \frac{u^4 + 2u^2 - u}{u (u^2 + 1)} du.$$

Polynomial long division yields the decomposition

$$\frac{u^4 + 2u^2 - u}{u (u^2 + 1)} = u + \frac{u^2 - u}{u (u^2 + 1)} = u + \frac{u - 1}{u^2 + 1}.$$

Continuing the integration process (on occasion using $v = u^2 + 1$, $dv = 2u du$)

$$\int \frac{u^4 + 2u^2 - u}{u (u^2 + 1)} du = \int \left(u + \frac{u - 1}{u^2 + 1} \right) du$$

$$= \frac{u^2}{2} + \frac{1}{2} \int \frac{2u}{u^2 + 1} du - \int \frac{1}{u^2 + 1} du$$

$$= \frac{u^2}{2} + \frac{1}{2} \int \frac{1}{v} dv - \arctan u$$

$$= \frac{u^2}{2} + \frac{1}{2} \ln |v| - \arctan u + C$$

$$= \frac{e^{2t}}{2} + \frac{1}{2} \ln (1 + e^{2t}) - \arctan (e^t) + C$$

Find the integral

$$\int \frac{2x - 3}{(x + 1)^2(x^2 + 4)} dx$$

Solution: Use partial fractions decomposition.

$$\frac{2x - 3}{(x + 1)^2(x^2 + 4)} = \frac{A}{x + 1} + \frac{B}{(x + 1)^2} + \frac{Cx + D}{x^2 + 4}$$

$$2x - 3 = A(x + 1)(x^2 + 4) + B(x^2 + 4) + (Cx + D)(x + 1)^2$$

$$2x - 3 = Ax^3 + 4Ax + Ax^2 + 4A + Bx^2 + 4B + Cx^3 + 2Cx^2 + Cx + Dx^2 + 2Dx + D$$

Identifying the coefficients according to the powers of x

$$x^3 : 0 = A + C$$

$$x^2 : 0 = A + B + 2C + D$$

$$x^1 : 2 = 4A + C + 2D$$

$$x^0 : -3 = 4A + 4B + D$$

$$A = C = 0, B = -1, D = 1$$

$$\int \frac{2x - 3}{(x + 1)^2(x^2 + 4)} dx = \int \left(-\frac{1}{(x + 1)^2} + \frac{1}{x^2 + 4} \right) dx = \frac{1}{x + 1} + \frac{1}{2} \arctan \frac{x}{2} + C$$

Sums and Sigma Notation

Sigma notation

If m and n are integers with $m \leq n$, and if f is a function defined at the integers $m, m + 1, m + 2, \dots, n$, the symbol $\sum_{i=m}^n f(i)$ represents the sum of the values of f at those integers:

$$\sum_{i=m}^n f(i) = f(m) + f(m + 1) + f(m + 2) + \cdots + f(n).$$

The explicit sum appearing on the right side of this equation is the **expansion** of the sum represented in sigma notation on the left side.

EXAMPLE 1

$$\sum_{i=1}^5 i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$$

The i that appears in the symbol $\sum_{i=m}^n f(i)$ is called an **index of summation**. To evaluate $\sum_{i=m}^n f(i)$, replace the index i with the integers $m, m + 1, \dots, n$, successively, and sum the results.

$$\sum_{k=1}^5 k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$$

The index of summation is a *dummy variable* used to represent an arbitrary point where the function is evaluated to produce a term to be included in the sum. On the other hand, the sum $\sum_{i=m}^n f(i)$ does depend on the two numbers m and n , called the **limits of summation**; m is the **lower limit**, and n is the **upper limit**.

EXAMPLE 2 (Examples of sums using sigma notation)

$$\sum_{j=1}^{20} j = 1 + 2 + 3 + \cdots + 18 + 19 + 20$$

$$\sum_{i=0}^n x^i = x^0 + x^1 + x^2 + \cdots + x^{n-1} + x^n$$

$$\sum_{m=1}^n 1 = \underbrace{1 + 1 + 1 + \cdots + 1}_{n \text{ terms}}$$

$$\sum_{k=-2}^3 \frac{1}{k+7} = \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

Sometimes we use a subscripted variable a_i to denote the i th term of a general sum instead of using the functional notation $f(i)$:

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + a_{m+2} + \cdots + a_n.$$

In particular, an **infinite series** is such a sum with infinitely many terms:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots.$$

When adding finitely many numbers, the order in which they are added is unimportant; any order will give the same sum. If all the numbers have a common factor, then that factor can be removed from each term and multiplied after the sum is evaluated: $ca + cb = c(a + b)$. These laws of arithmetic translate into the following *linearity* rule for finite sums; if A and B are constants, then

$$\sum_{i=m}^n (Af(i) + Bg(i)) = A \sum_{i=m}^n f(i) + B \sum_{i=m}^n g(i).$$

Both of the sums $\sum_{j=m}^{m+n} f(j)$ and $\sum_{i=0}^n f(i + m)$ have the same expansion, namely, $f(m) + f(m + 1) + \cdots + f(m + n)$. Therefore, the two sums are equal.

$$\sum_{j=m}^{m+n} f(j) = \sum_{i=0}^n f(i + m).$$

This equality can also be derived by substituting $i + m$ for j everywhere j appears on the left side, noting that $i + m = m$ reduces to $i = 0$, and $i + m = m + n$ reduces to $i = n$. It is often convenient to make such a **change of index** in a summation.

EXAMPLE 3 Express $\sum_{j=3}^{17} \sqrt{1 + j^2}$ in the form $\sum_{i=1}^n f(i)$.

Solution Let $j = i + 2$. Then $j = 3$ corresponds to $i = 1$ and $j = 17$ corresponds to $i = 15$. Thus,

$$\sum_{j=3}^{17} \sqrt{1 + j^2} = \sum_{i=1}^{15} \sqrt{1 + (i + 2)^2}.$$

Evaluating Sums

There is a **closed form** expression for the sum S of the first n positive integers, namely,

$$S = \sum_{i=1}^n i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

To see this, write the sum forwards and backwards and add the two to get

$$\begin{array}{rcccccccc} S = & 1 & + & 2 & + & 3 & + \cdots + (n-1) & + & n \\ S = & n & + & (n-1) & + & (n-2) & + \cdots + 2 & + & 1 \\ \hline 2S = & (n+1) & + & (n+1) & + & (n+1) & + \cdots + (n+1) & + & (n+1) = n(n+1) \end{array}$$

The formula for S follows when we divide by 2.

It is not usually this easy to evaluate a general sum in closed form. We can only simplify $\sum_{i=m}^n f(i)$ for a small class of functions f .

THEOREM Summation formulas

$$(a) \quad \sum_{i=1}^n 1 = \underbrace{1 + 1 + 1 + \cdots + 1}_{n \text{ terms}} = n.$$

$$(b) \quad \sum_{i=1}^n i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

$$(c) \quad \sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$(d) \quad \sum_{i=1}^n r^{i-1} = 1 + r + r^2 + r^3 + \cdots + r^{n-1} = \frac{r^n - 1}{r - 1} \quad \text{if } r \neq 1.$$

Evaluate $\sum_{k=m+1}^n (6k^2 - 4k + 3)$, where $1 \leq m < n$.

Solution Using the rules of summation and various summation formulas from Theorem we calculate

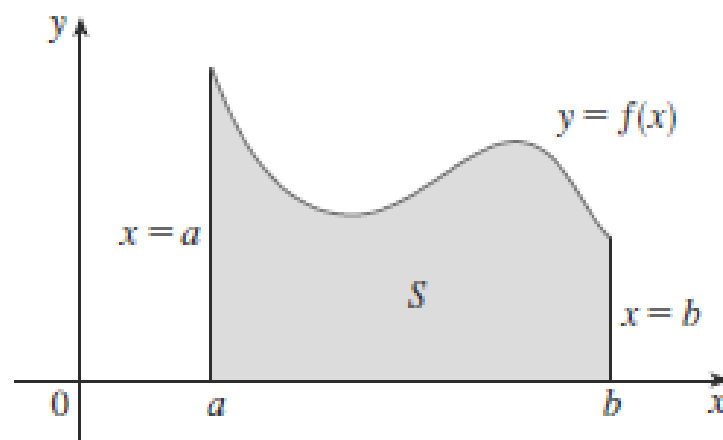
$$\begin{aligned}\sum_{k=1}^n (6k^2 - 4k + 3) &= 6 \sum_{k=1}^n k^2 - 4 \sum_{k=1}^n k + 3 \sum_{k=1}^n 1 \\ &= 6 \frac{n(n+1)(2n+1)}{6} - 4 \frac{n(n+1)}{2} + 3n \\ &= 2n^3 + n^2 + 2n\end{aligned}$$

Thus,

$$\begin{aligned}\sum_{k=m+1}^n (6k^2 - 4k + 3) &= \sum_{k=1}^n (6k^2 - 4k + 3) - \sum_{k=1}^m (6k^2 - 4k + 3) \\ &= 2n^3 + n^2 + 2n - 2m^3 - m^2 - 2m.\end{aligned}$$

The Area Problem

We begin by attempting to solve the *area problem*: Find the area of the region S that lies under the curve $y = f(x)$ from a to b . This means that S , is bounded by the graph of a continuous function f [where $f(x) \geq 0$], the vertical lines $x = a$ and $x = b$, and the x -axis.



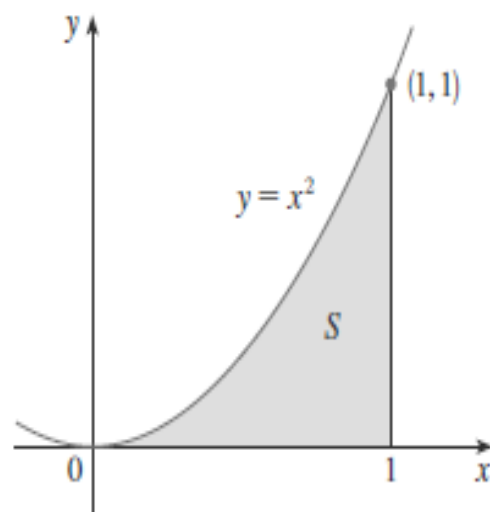
$$S = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}$$

FIGURE 1

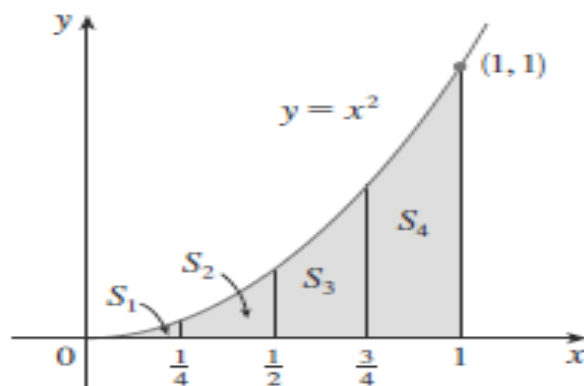
However, it isn't so easy to find the area of a region with curved sides. We all have an intuitive idea of what the area of a region is. But part of the area problem is to make this intuitive idea precise by giving an exact definition of area.

Recall that in defining a tangent we first approximated the slope of the tangent line by slopes of secant lines and then we took the limit of these approximations. We pursue a similar idea for areas. We first approximate the region S by rectangles and then we take the limit of the areas of these rectangles as we increase the number of rectangles. The following example illustrates the procedure.

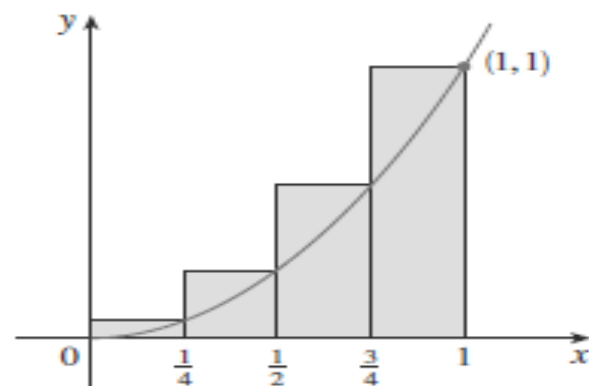
EXAMPLE 1 Use rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 (the parabolic region S illustrated in Figure



SOLUTION We first notice that the area of S must be somewhere between 0 and 1 because S is contained in a square with side length 1, but we can certainly do better than that. Suppose we divide S into four strips S_1 , S_2 , S_3 , and S_4 by drawing the vertical lines $x = \frac{1}{4}$, $x = \frac{1}{2}$, and $x = \frac{3}{4}$ as in Figure (a).



(a)



(b)

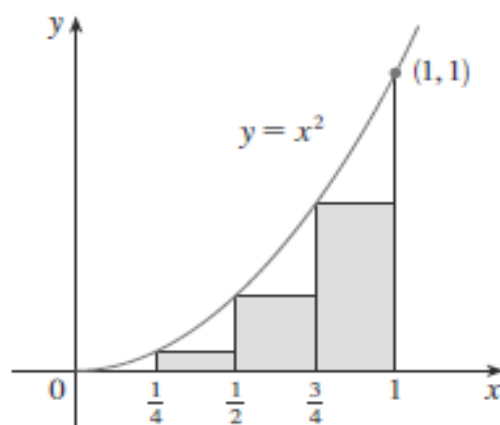
We can approximate each strip by a rectangle whose base is the same as the strip and whose height is the same as the right edge of the strip [see Figure (b)]. In other words, the heights of these rectangles are the values of the function $f(x) = x^2$ at the right endpoints of the subintervals $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$, and $[\frac{3}{4}, 1]$.

Each rectangle has width $\frac{1}{4}$ and the heights are $(\frac{1}{4})^2$, $(\frac{1}{2})^2$, $(\frac{3}{4})^2$, and 1^2 . If we let R_4 be the sum of the areas of these approximating rectangles, we get

$$R_4 = \frac{1}{4} \cdot \left(\frac{1}{4}\right)^2 + \frac{1}{4} \cdot \left(\frac{1}{2}\right)^2 + \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 + \frac{1}{4} \cdot 1^2 = \frac{15}{32} = 0.46875$$

From Figure (b) we see that the area A of S is less than R_4 , so

$$A < 0.46875$$



Instead of using the rectangles in Figure (b) we could use the smaller rectangles in Figure whose heights are the values of f at the left endpoints of the subintervals. (The leftmost rectangle has collapsed because its height is 0.) The sum of the areas of these approximating rectangles is

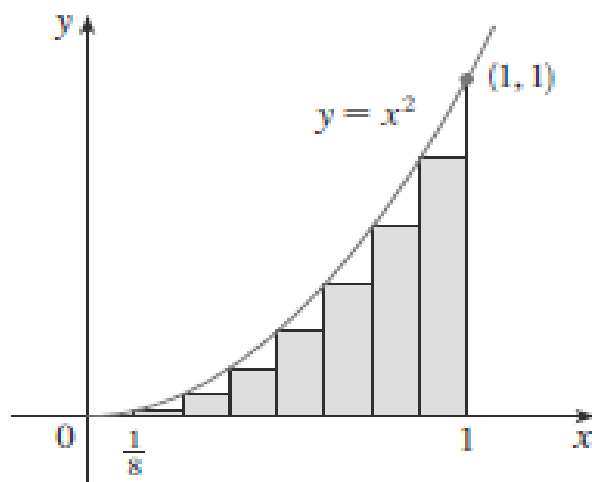
$$L_4 = \frac{1}{4} \cdot 0^2 + \frac{1}{4} \cdot \left(\frac{1}{4}\right)^2 + \frac{1}{4} \cdot \left(\frac{1}{2}\right)^2 + \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 = \frac{7}{32} = 0.21875$$

We see that the area of S is larger than L_4 , so we have lower and upper estimates for A :

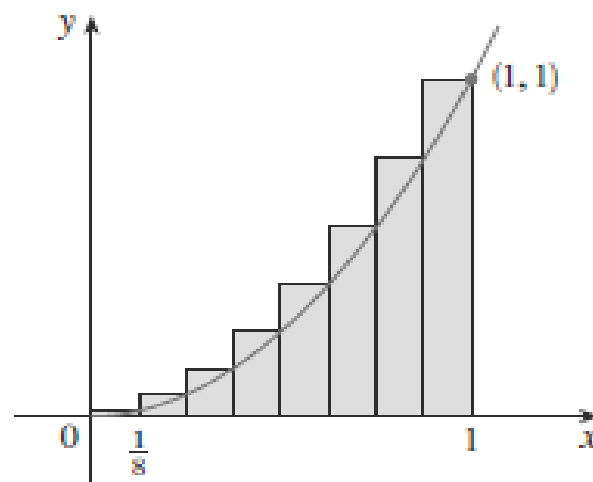
$$0.21875 < A < 0.46875$$

Following

We can repeat this procedure with a larger number of strips. Figure shows what happens when we divide the region S into eight strips of equal width.



(a) Using left endpoints



(b) Using right endpoints

By computing the sum of the areas of the smaller rectangles (L_8) and the sum of the areas of the larger rectangles (R_8), we obtain better lower and upper estimates for A :

$$0.2734375 < A < 0.3984375$$

So one possible answer to the question is to say that the true area of S lies somewhere between 0.2734375 and 0.3984375.

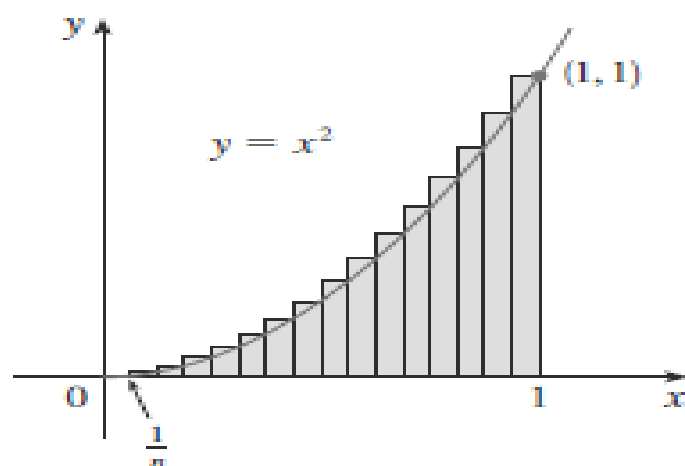
We could obtain better estimates by increasing the number of strips.

EXAMPLE 2 For the region S in Example 1, show that the sum of the areas of the upper approximating rectangles approaches $\frac{1}{3}$, that is,

$$\lim_{n \rightarrow \infty} R_n = \frac{1}{3}$$

SOLUTION R_n is the sum of the areas of the n rectangles in Figure 1. Each rectangle has width $1/n$ and the heights are the values of the function $f(x) = x^2$ at the points $1/n, 2/n, 3/n, \dots, n/n$; that is, the heights are $(1/n)^2, (2/n)^2, (3/n)^2, \dots, (n/n)^2$.

Thus



$$\begin{aligned} R_n &= \frac{1}{n} \left(\frac{1}{n} \right)^2 + \frac{1}{n} \left(\frac{2}{n} \right)^2 + \frac{1}{n} \left(\frac{3}{n} \right)^2 + \cdots + \frac{1}{n} \left(\frac{n}{n} \right)^2 \\ &= \frac{1}{n} \cdot \frac{1}{n^2} (1^2 + 2^2 + 3^2 + \cdots + n^2) \\ &= \frac{1}{n^3} (1^2 + 2^2 + 3^2 + \cdots + n^2) \end{aligned}$$

Here we need the formula for the sum of the squares of the first n positive integers:

$$\boxed{1} \quad 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Putting Formula 1 into our expression for R_n , we get

$$R_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

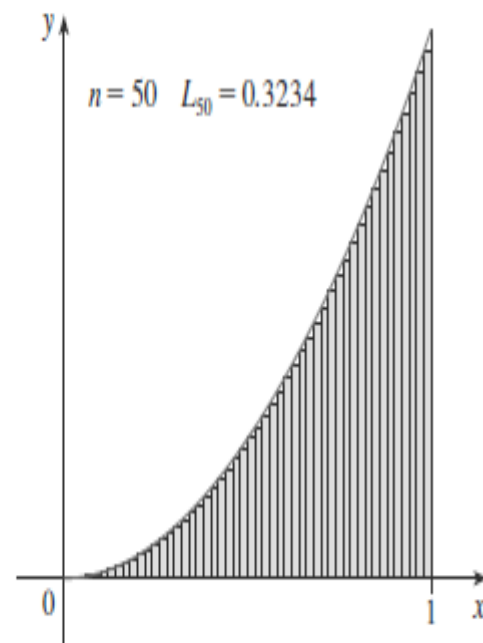
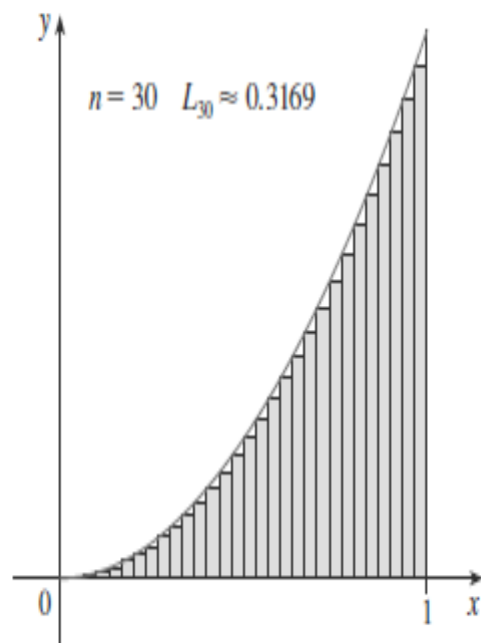
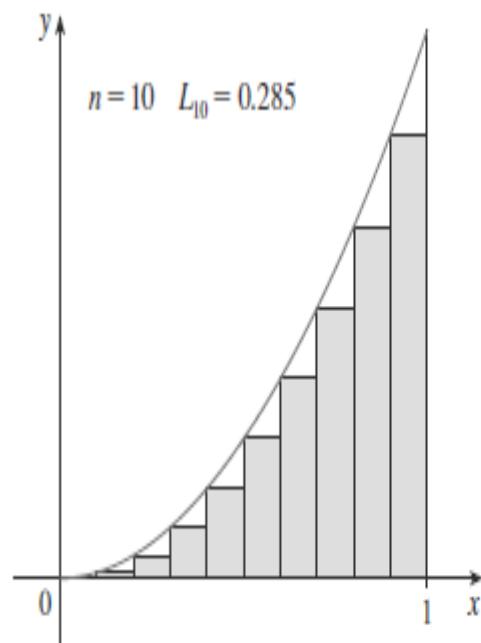
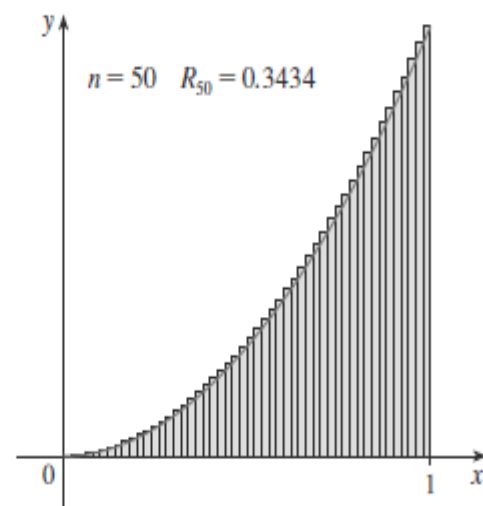
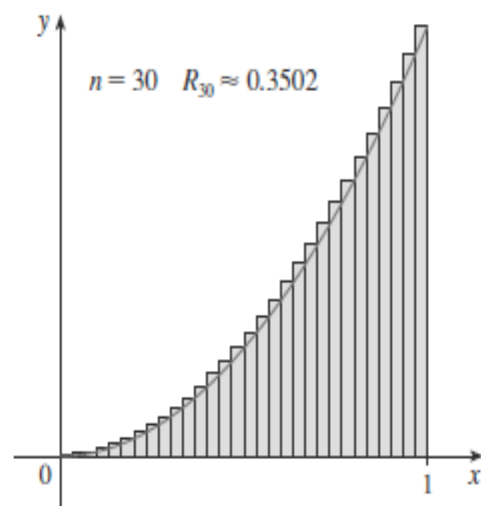
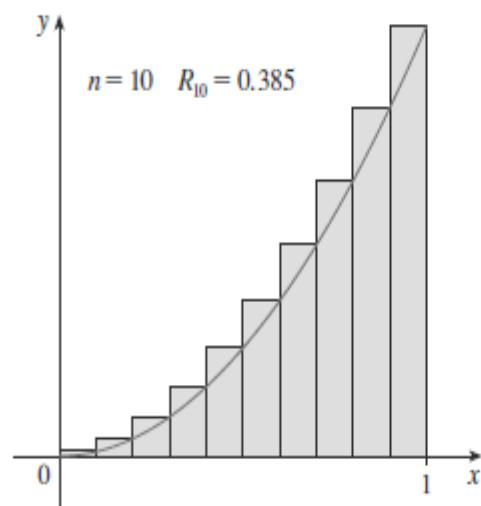
Thus, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} R_n &= \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)}{6n^2} \\ &= \lim_{n \rightarrow \infty} \frac{1}{6} \left(\frac{n+1}{n} \right) \left(\frac{2n+1}{n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \\ &= \frac{1}{6} \cdot 1 \cdot 2 = \frac{1}{3} \end{aligned}$$

It can be shown that the lower approximating sums also approach $\frac{1}{3}$, that is,

$$\lim_{n \rightarrow \infty} L_n = \frac{1}{3}$$

From Figures it appears that, as n increases, both L_n and R_n become better and better approximations to the area of S .



Therefore, we *define* the area A to be the limit of the sums of the areas of the approximating rectangles, that is,

$$A = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} L_n = \frac{1}{3}$$

Let's apply the idea of Examples 1 and 2 to the more general region S of Figure 1. We start by subdividing S into n strips $S_1, S_2, \dots, S_n$ of equal width as in Figure . The width of the interval $[a, b]$ is $b - a$, so the width of each of the n strips is

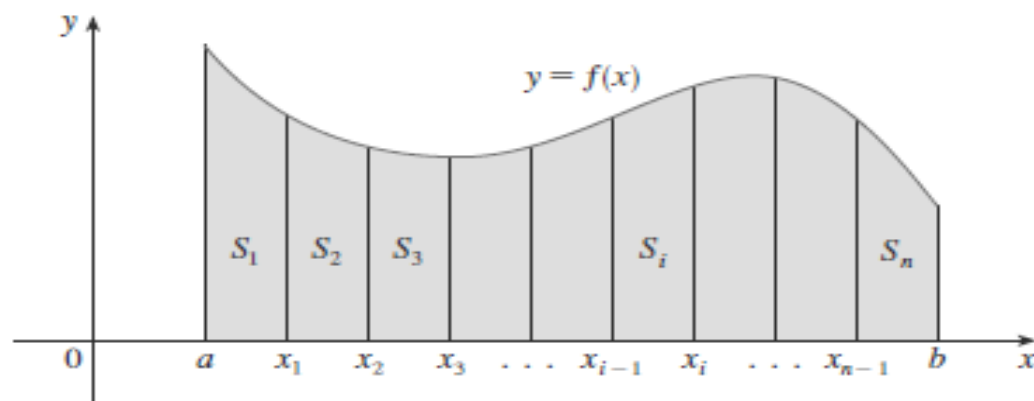
$$\Delta x = \frac{b - a}{n}$$

These strips divide the interval $[a, b]$ into n subintervals

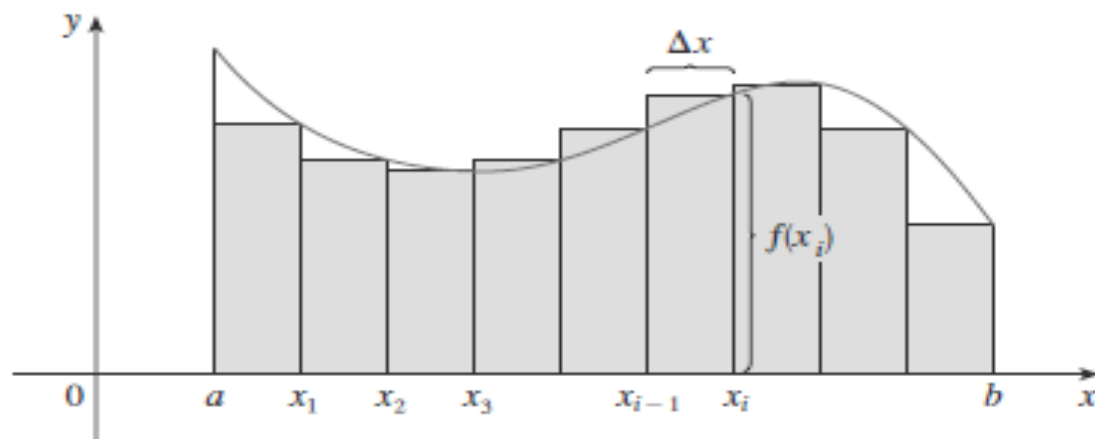
$$[x_0, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, x_n]$$

where $x_0 = a$ and $x_n = b$. The right endpoints of the subintervals are

$$x_1 = a + \Delta x, \quad x_2 = a + 2\Delta x, \quad x_3 = a + 3\Delta x, \quad \dots$$

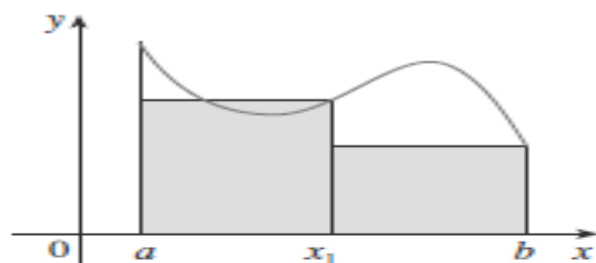


Let's approximate the i th strip S_i by a rectangle with width Δx and height $f(x_i)$, which is the value of f at the right endpoint (see Figure). Then the area of the i th rectangle

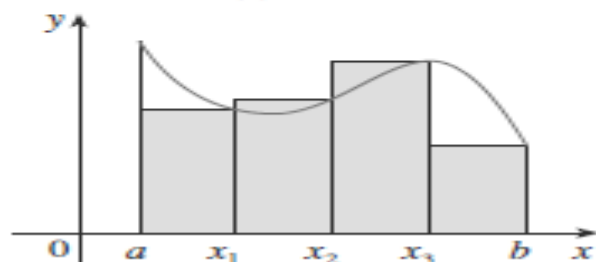


is $f(x_i) \Delta x$. What we think of intuitively as the area of S is approximated by the sum of the areas of these rectangles, which is

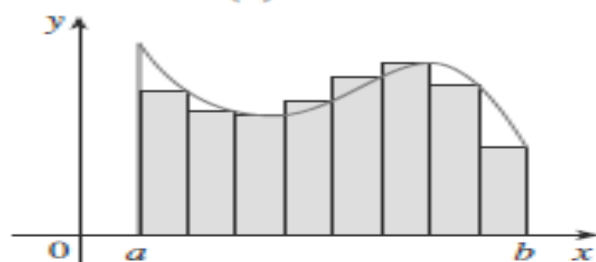
$$R_n = f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x$$



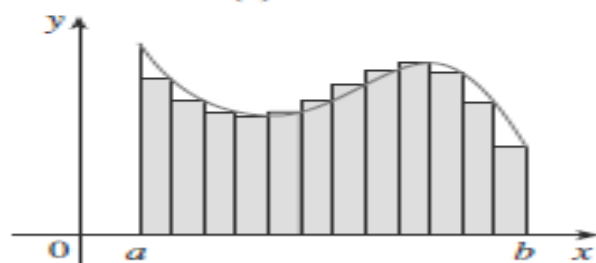
(a) $n = 2$



(b) $n = 4$



(c) $n = 8$



(d) $n = 12$

Figure shows this approximation for $n = 2, 4, 8$, and 12 . Notice that this approximation appears to become better and better as the number of strips increases, that is, as $n \rightarrow \infty$. Therefore, we define the area A of the region S in the following way.

2 Definition The area A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles:

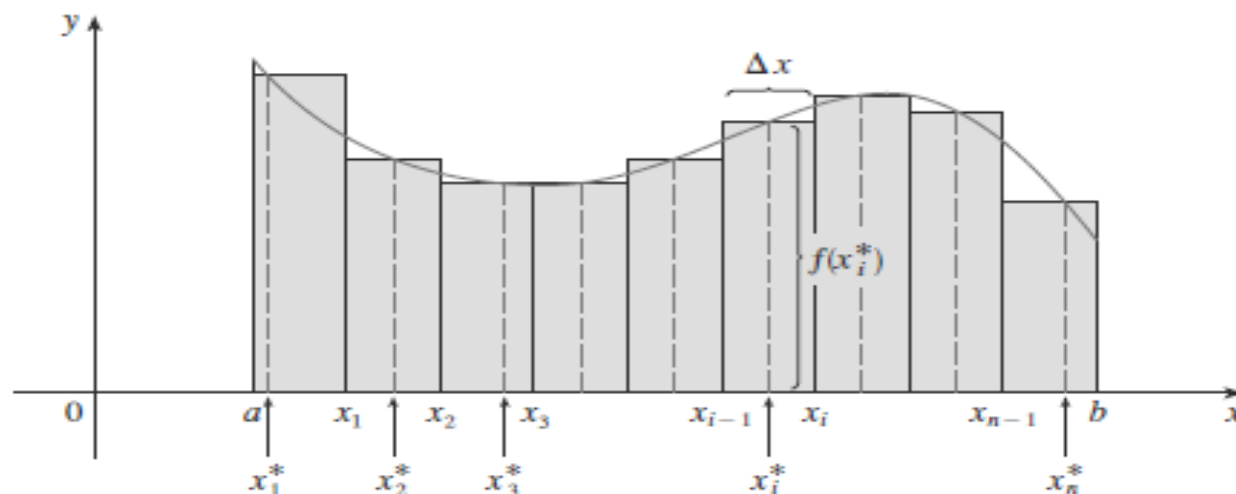
$$A = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} [f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x]$$

It can be proved that the limit in Definition 2 always exists, since we are assuming that f is continuous. It can also be shown that we get the same value if we use left endpoints:

$$\boxed{3} \quad A = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} [f(x_0) \Delta x + f(x_1) \Delta x + \cdots + f(x_{n-1}) \Delta x]$$

In fact, instead of using left endpoints or right endpoints, we could take the height of the i th rectangle to be the value of f at *any* number x_i^* in the i th subinterval $[x_{i-1}, x_i]$. We call the numbers $x_1^*, x_2^*, \dots, x_n^*$ the **sample points**. Figure 5.1 shows approximating rectangles when the sample points are not chosen to be endpoints. So a more general expression for the area of S is

$$\boxed{4} \quad A = \lim_{n \rightarrow \infty} [f(x_1^*) \Delta x + f(x_2^*) \Delta x + \cdots + f(x_n^*) \Delta x]$$



We often use **sigma notation** to write sums with many terms more compactly. For instance,

$$\sum_{i=1}^n f(x_i) \Delta x = f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x$$

This tells us to
end with $i = n$.

This tells us
to add.

This tells us to
start with $i = m$.

$$\sum_{i=m}^n f(x_i) \Delta x$$

So the expressions for area in Equations 2, 3, and 4 can be written as follows:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

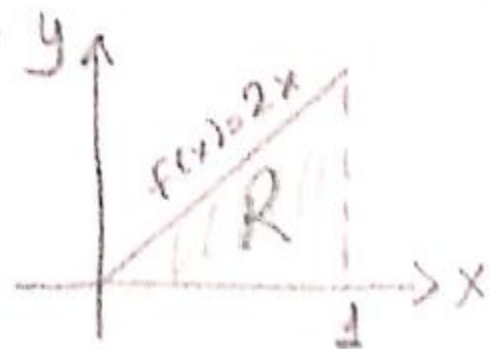
$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_{i-1}) \Delta x$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

We could also rewrite Formula 1 in the following way:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

①



$f(x) = 2x$; R is bounded by
 $f(x) = 2x$, x-axis and $x = 1$

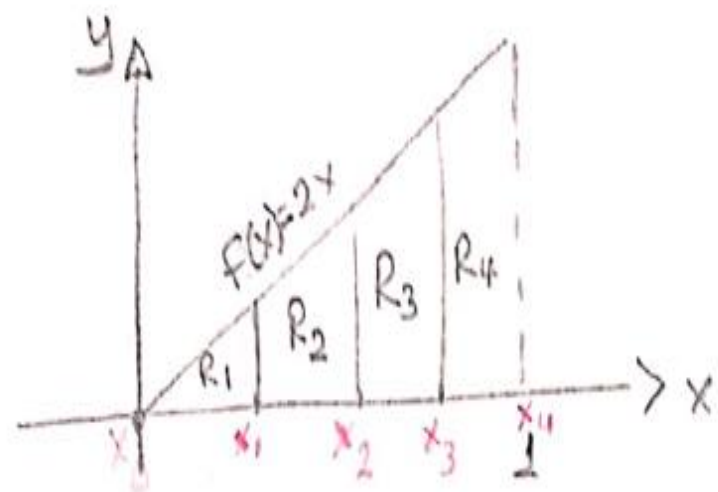
R is a right triangle

b: base

h: height

$$A = \frac{1}{2} \cdot b \cdot h = \frac{1}{2} \cdot 1 \cdot 2 = 1$$

Divide $[0,1]$ into four subintervals of equal length
for $n=4$:



$$x_0 = 0$$

$$x_1 = 1/4$$

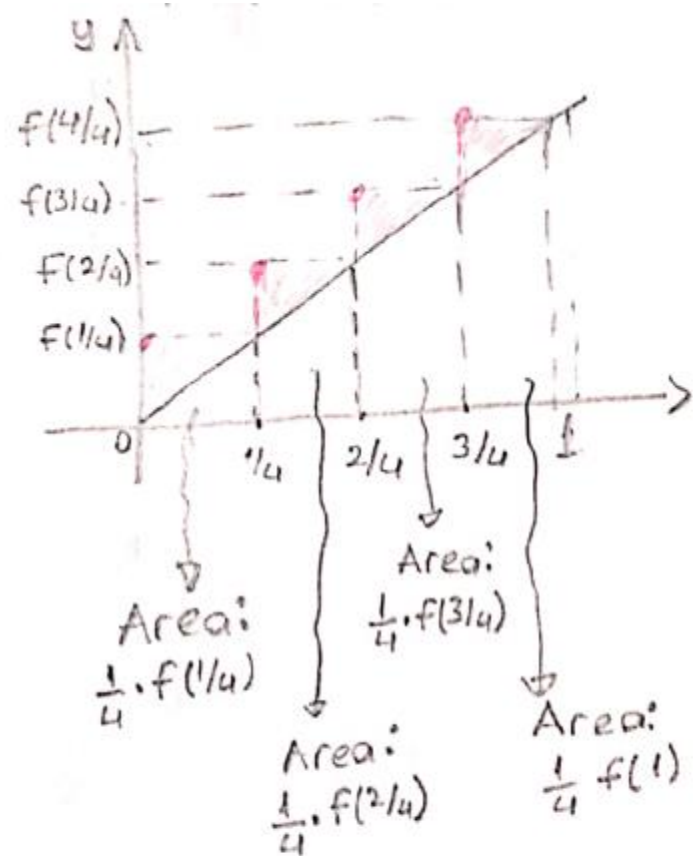
$$x_2 = 2/4$$

$$x_3 = 3/4$$

$$x_4 = 4/4 = 1$$

$\Delta x = \frac{1}{4}$ length of subintervals

R_1, R_2, R_3, R_4 subregions of R



$$f(x) = y = 2x$$

$$f(1/4) = 2 \cdot 1/4$$

$$f(2/4) = 2 \cdot 2/4$$

$$f(3/4) = 2 \cdot 3/4$$

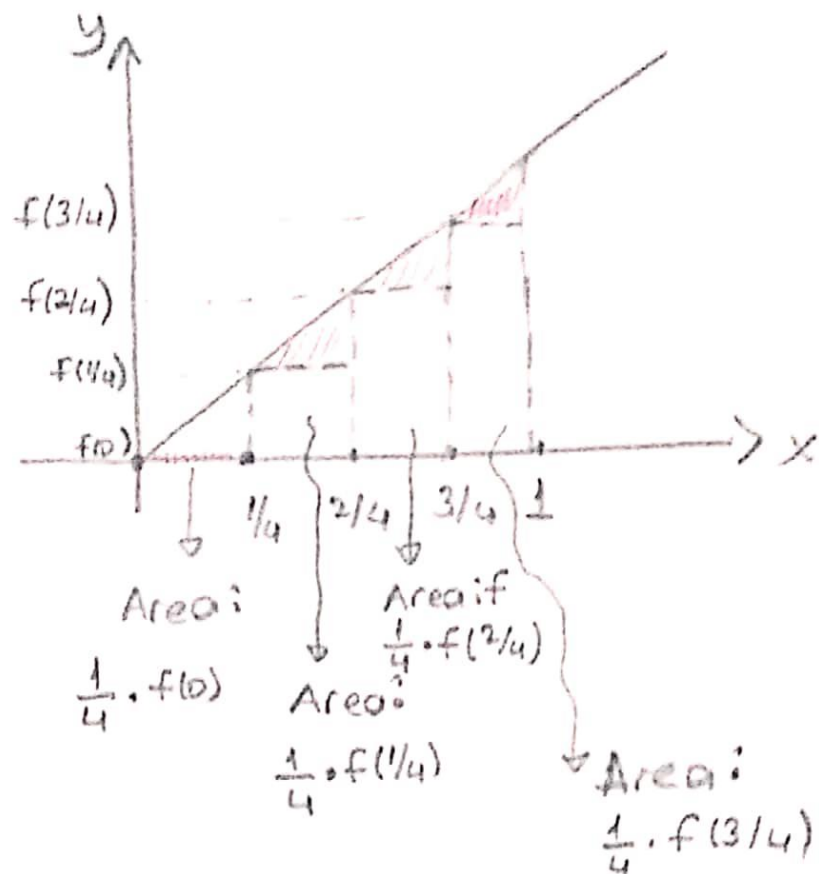
$$f(4/4) = 2 \cdot 4/4$$

Sum of these areas:

$$\overline{S}_4 = \frac{1}{4} [2 \cdot 1/4 + 2 \cdot 2/4 + 2 \cdot 3/4 + 2 \cdot 1] = \frac{5}{4}$$

$$\overline{S}_4 = \sum_{i=1}^4 f(x_i) \cdot \Delta x$$

$\overline{S}_4 >$ area of the triangle



$$f(x) = 2x$$

$$f(0) = 0$$

$$f(1/4) = 2/4$$

$$f(2/4) = 2 \cdot 2/4 = 1$$

$$f(3/4) = 2 \cdot 3/4 = 3/2$$

(2)

$$S_4 = \frac{1}{4} [f(0) + f(1/4) + f(2/4) + f(3/4)]$$

$$= \frac{1}{4} [0 + 1/2 + 1 + 3/2] = 3/4$$

$$\underline{S_4} = \sum_{i=1}^3 f(x_i) \cdot \Delta x_i$$

$\underline{S_4} < \text{the area of triangle}$

$$\text{So; } \underline{S_4} \leq A \leq \overline{S_4}$$

More generally, if we divide $[0,1]$ into n subintervals of equal length Δx , then

$$\Delta x = \frac{1}{n}$$

$$\overline{S_n} = \sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \Delta x = \frac{n+1}{n}$$

$$\underline{S_n} = \sum_{k=1}^{n-1} f\left(\frac{k}{n}\right) \cdot \Delta x = \frac{n-1}{n}$$

$\Rightarrow$ As n becomes larger
 $\underline{S_n}$ and $\overline{S_n}$
 become better approximation to A .

$$\overline{S}_n = \sum_{k=1}^n f\left(\frac{k}{n}\right) \cdot \Delta x$$

$$f(x) = y = 2x$$

$$f\left(\frac{k}{n}\right) = 2 \cdot \frac{k}{n} \quad \Delta x = \frac{1}{n}$$

$$= \sum_{k=1}^n \frac{2k}{n^2}$$

$$= \frac{2}{n^2} \sum_{k=1}^n k = \frac{2}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{n}$$

$$\underline{S}_n = \sum_{k=1}^n f\left(\frac{k-1}{n}\right) \cdot \Delta x$$

$$= \sum_{k=1}^n 2\left(\frac{k-1}{n}\right) \cdot \frac{1}{n}$$

$$= \frac{2}{n^2} \sum_{k=1}^n (k-1)$$

$$= \frac{2}{n^2} \sum_{k=1}^{n-1} k$$

$$= \frac{2}{n^2} \cdot \frac{(n-1) \cdot n}{2} = \frac{n-1}{n}$$

$$\left. \begin{aligned} \lim_{n \rightarrow \infty} \underline{S}_n &= \lim_{n \rightarrow \infty} \frac{n-1}{n} = 1 \\ \lim_{n \rightarrow \infty} \overline{S}_n &= \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1 \end{aligned} \right\} \Rightarrow A=1$$

The common limit of $\overline{S}_n$ and $\underline{S}_n$ as $n \rightarrow \infty$ if it exists, is called the definite integral of f over $[a, b]$ and is written

$$\int_a^b f(x) dx$$

a : lower limit

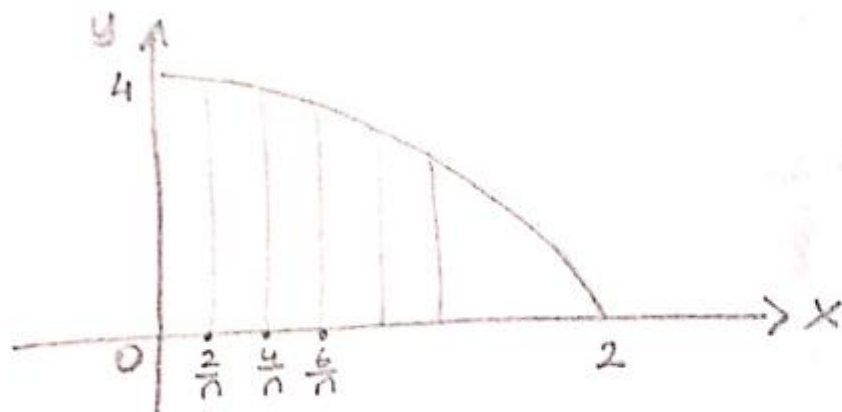
b : upper limit

That is,

$$\lim_{n \rightarrow \infty} \sum_{i=0}^n f(x_i) \Delta x_i = \int_a^b f(x) dx$$

Ex: Find the area of the region in the 1st quadrant bounded by $f(x) = 4 - x^2$ and the lines $x=0$ and $y=0$

$$\Delta x = \frac{2}{n}$$



$$\begin{aligned} \sum_{i=1}^n f(x_i) \cdot \Delta x &= f(x_1) \cdot \Delta x + f(x_2) \cdot \Delta x + \dots + f(x_n) \cdot \Delta x \\ &= f\left(\frac{2}{n}\right) \cdot \frac{2}{n} + f\left(\frac{4}{n}\right) \cdot \frac{2}{n} + \dots + f\left(\frac{2n}{n}\right) \cdot \frac{2}{n} \\ &= \frac{2}{n} \left\{ \left[\left(4 - \left(\frac{2}{n}\right)^2\right) + \left(4 - \left(\frac{4}{n}\right)^2\right) + \dots + \left(4 - \left(\frac{2n}{n}\right)^2\right) \right] \right\} \\ &= \frac{2}{n} \left[4n - \left(\frac{2}{n}\right)^2 \{1^2 + 2^2 + \dots + n^2\} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{n} \left[4n - \left(\frac{2}{n} \right)^2 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} \right] \\
&= \frac{2}{\cancel{n}} \cdot 4\cancel{n} - \frac{2}{\cancel{n}} \left(\frac{2}{n} \right)^2 \left\{ \frac{\cancel{n}(n+1)(2n+1)}{6} \right\} \\
&= 8 - \frac{8}{6n^2} \cdot (2n^2 + 3n + 1)
\end{aligned}$$

$$\lim_{n \rightarrow \infty} S_n = 8 - \frac{8}{3} = \frac{16}{3}$$

Evaluate the following Definite integral by using Riemann sums.

$$\int_0^3 (9 - x^2) dx$$

For the right edge of equal length subintervals

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_i = a + i\Delta x, \quad i = 0 \dots n.$$

$$\Delta x = \frac{3-0}{n} = \frac{3}{n}, \quad x_i = 0 + i\frac{3}{n} = \frac{3i}{n}, \quad i = 0 \dots n. \quad f(x_i) = 9 - \left(\frac{3i}{n}\right)^2$$

$$\begin{aligned} \int_0^3 (9 - x^2) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(9 - \frac{9i^2}{n^2} \right) \frac{3}{n} = \lim_{n \rightarrow \infty} \left[\frac{27}{n} \sum_{i=1}^n \left(1 - \frac{i^2}{n^2} \right) \right] = \lim_{n \rightarrow \infty} \left[\frac{27n}{n} - \frac{27}{n^3} \sum_{i=1}^n i^2 \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{27n}{n} - \frac{27}{n^3} \frac{n(n+1)(2n+1)}{6} \right] = 27 - 27 \left(\frac{1}{3} \right) = 18. \end{aligned}$$

Let A be the area of the region that lies under the graph of $f(x) = e^{-x}$ between $x = 0$ and $x = 2$.

(a) Using right endpoints, find an expression for A as a limit. Do not evaluate the limit.

(b) Estimate the area by taking the sample points to be midpoints and using four subintervals and then ten subintervals.

(a) Since $a = 0$ and $b = 2$, the width of a subinterval is

$$\Delta x = \frac{2 - 0}{n} = \frac{2}{n}$$

So $x_1 = 2/n$, $x_2 = 4/n$, $x_3 = 6/n$, $x_i = 2i/n$, and $x_n = 2n/n$. The sum of the areas of the approximating rectangles is

$$\begin{aligned} R_n &= f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x \\ &= e^{-x_1} \Delta x + e^{-x_2} \Delta x + \cdots + e^{-x_n} \Delta x \\ &= e^{-2/n} \left(\frac{2}{n} \right) + e^{-4/n} \left(\frac{2}{n} \right) + \cdots + e^{-2n/n} \left(\frac{2}{n} \right) \end{aligned}$$

According to Definition 2, the area is

$$A = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{2}{n} (e^{-2/n} + e^{-4/n} + e^{-6/n} + \cdots + e^{-2n/n})$$

Using sigma notation we could write

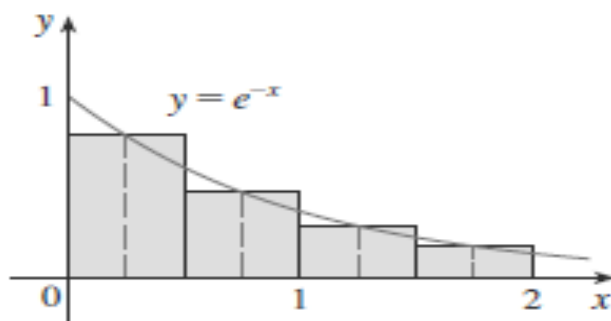
$$A = \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n e^{-2i/n}$$

(b) With $n = 4$ the subintervals of equal width $\Delta x = 0.5$ are $[0, 0.5]$, $[0.5, 1]$, $[1, 1.5]$, and $[1.5, 2]$. The midpoints of these subintervals are $x_1^* = 0.25$, $x_2^* = 0.75$, $x_3^* = 1.25$, and $x_4^* = 1.75$, and the sum of the areas of the four approximating rectangles is

$$\begin{aligned} M_4 &= \sum_{i=1}^4 f(x_i^*) \Delta x \\ &= f(0.25) \Delta x + f(0.75) \Delta x + f(1.25) \Delta x + f(1.75) \Delta x \\ &= e^{-0.25}(0.5) + e^{-0.75}(0.5) + e^{-1.25}(0.5) + e^{-1.75}(0.5) \\ &= \frac{1}{2}(e^{-0.25} + e^{-0.75} + e^{-1.25} + e^{-1.75}) \approx 0.8557 \end{aligned}$$

So an estimate for the area is

$$A \approx 0.8557$$



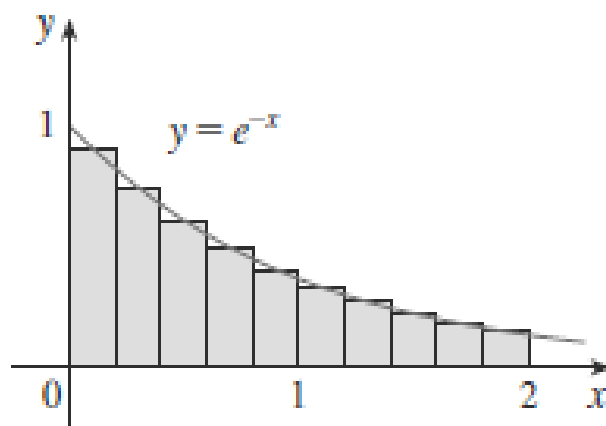
So an estimate for the area is

$$A \approx 0.8557$$

With $n = 10$ the subintervals are $[0, 0.2], [0.2, 0.4], \dots, [1.8, 2]$ and the midpoints are $x_1^* = 0.1, x_2^* = 0.3, x_3^* = 0.5, \dots, x_{10}^* = 1.9$. Thus

$$\begin{aligned} A &\approx M_{10} = f(0.1) \Delta x + f(0.3) \Delta x + f(0.5) \Delta x + \cdots + f(1.9) \Delta x \\ &= 0.2(e^{-0.1} + e^{-0.3} + e^{-0.5} + \cdots + e^{-1.9}) \approx 0.8632 \end{aligned}$$

From Figure it appears that this estimate is better than the estimate with $n = 4$.



Homework

Evaluate the following limit

$$A = \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n e^{-2i/n}$$

The Definite Integral

$$\boxed{1} \quad \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} [f(x_1^*) \Delta x + f(x_2^*) \Delta x + \cdots + f(x_n^*) \Delta x]$$

arises when we compute an area.

2 Definition of a Definite Integral If f is a continuous function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a)$, $x_1, x_2, \dots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$. Then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

Because we have assumed that f is continuous, it can be proved that the limit in Definition 2 always exists and gives the same value no matter how we choose the sample points x_i^* .

If we take the sample points to be right endpoints, then $x_i^* = x_i$ and the definition of an integral becomes

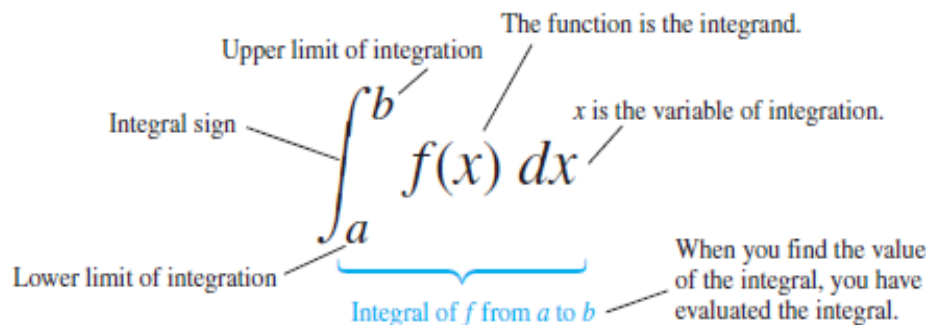
$$\boxed{3} \quad \int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$

If we choose the sample points to be left endpoints, then $x_i^* = x_{i-1}$ and the definition becomes

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_{i-1}) \Delta x$$

Alternatively, we could choose x_i^* to be the midpoint of the subinterval or any other number between x_{i-1} and x_i .

$$\int_a^b f(x) dx = \int_a^b f(t) dt.$$



The various parts of the symbol $\int_a^b f(x) dx$ have their own names:

- (i) $\int$ is called the **integral sign**; it resembles the letter S since it represents the limit of a sum.
- (ii) a and b are called the **limits of integration**; a is the **lower limit**, b is the **upper limit**.
- (iii) The function f is the **integrand**; x is the **variable of integration**.
- (iv) dx is the **differential** of x . It replaces Δx in the Riemann sums. If an integrand depends on more than one variable, the differential tells you which one is the variable of integration.

NOTE 1 • The symbol $\int$ was introduced by Leibniz and is called an **integral sign**. It is an elongated S and was chosen because an integral is a limit of sums. In the notation $\int_a^b f(x) dx$, $f(x)$ is called the **integrand** and a and b are called the **limits of integration**; a is the **lower limit** and b is the **upper limit**. The symbol dx has no official meaning by itself; $\int_a^b f(x) dx$ is all one symbol. The procedure of calculating an integral is called **integration**.

NOTE 2 • The definite integral $\int_a^b f(x) dx$ is a number; it does not depend on x . In fact, we could use any letter in place of x without changing the value of the integral:

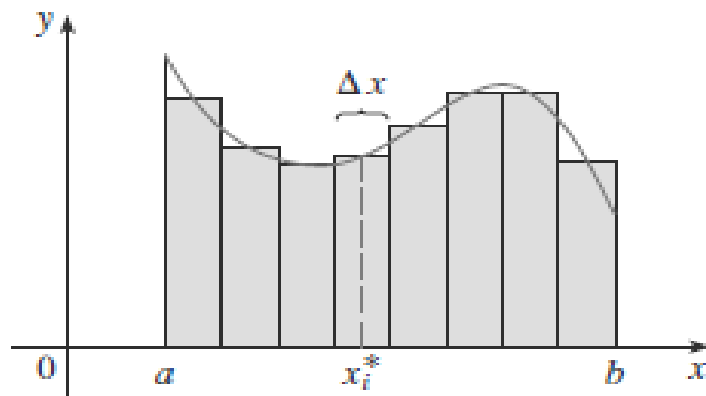
$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(r) dr$$

NOTE 3 • The sum

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

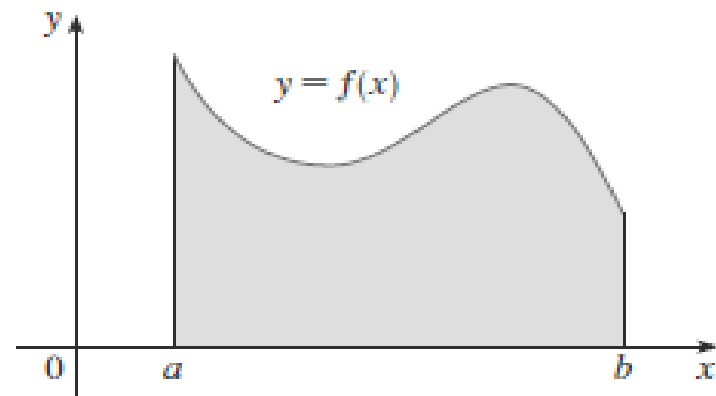
that occurs in Definition 2 is called a **Riemann sum** after the German mathematician Bernhard Riemann (1826–1866). We know that if f happens to be positive, then the Riemann sum can be interpreted as a sum of areas of approximating rectangles

the definite integral $\int_a^b f(x) dx$ can be interpreted as the area under the curve $y = f(x)$ from a to b .



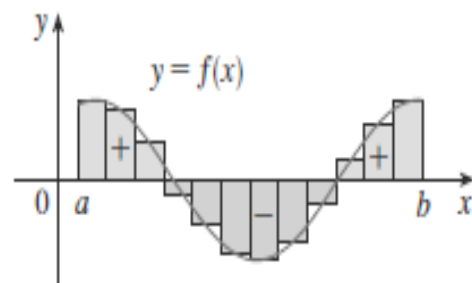
|

If $f(x) \geq 0$, the Riemann sum $\sum f(x_i^*) \Delta x$ is the sum of areas of rectangles.

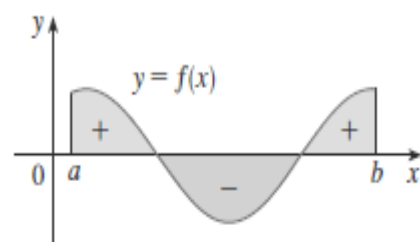


|

If $f(x) \geq 0$, the integral $\int_a^b f(x) dx$ is the area under the curve $y = f(x)$ from a to b .



$\sum f(x_i^*) \Delta x$ is an approximation to the net area



$\int_a^b f(x) dx$ is the net area

$$\int_a^b f(x) dx = A_1 - A_2$$

where A_1 is the area of the region above the x -axis and below the graph of f , and A_2 is the area of the region below the x -axis and above the graph of f .

EXAMPLE 1 Express

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \Delta x$$

as an integral on the interval $[0, \pi]$.

SOLUTION Comparing the given limit with the limit in Definition 2, we see that they will be identical if we choose

$$f(x) = x^3 + x \sin x \quad \text{and} \quad x_i^* = x_i$$

(So the sample points are right endpoints and the given limit is of the form of Equation 3.) We are given that $a = 0$ and $b = \pi$.

we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (x_i^3 + x_i \sin x_i) \Delta x = \int_0^{\pi} (x^3 + x \sin x) dx$$

LET US REMEMBER

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

The remaining formulas are simple rules for working with sigma notation:

$$\sum_{i=1}^n c = nc$$

$$\sum_{i=1}^n ca_i = c \sum_{i=1}^n a_i$$

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

$$\sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i$$

EXAMPLE

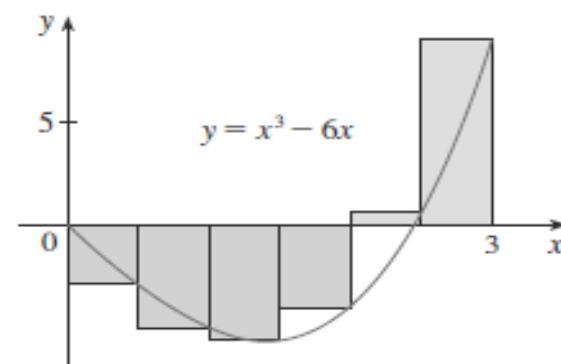
(a) Evaluate the Riemann sum for $f(x) = x^3 - 6x$ taking the sample points to be right endpoints and $a = 0$, $b = 3$, and $n = 6$.

(b) Evaluate $\int_0^3 (x^3 - 6x) dx$.

SOLUTION

(a) With $n = 6$ the interval width is

$$\Delta x = \frac{b - a}{n} = \frac{3 - 0}{6} = \frac{1}{2}$$



and the right endpoints are $x_1 = 0.5$, $x_2 = 1.0$, $x_3 = 1.5$, $x_4 = 2.0$, $x_5 = 2.5$, and $x_6 = 3.0$. So the Riemann sum is

$$\begin{aligned} R_6 &= \sum_{i=1}^6 f(x_i) \Delta x \\ &= f(0.5) \Delta x + f(1.0) \Delta x + f(1.5) \Delta x + f(2.0) \Delta x + f(2.5) \Delta x + f(3.0) \Delta x \\ &= \frac{1}{2}(-2.875 - 5 - 5.625 - 4 + 0.625 + 9) \\ &= -3.9375 \end{aligned}$$

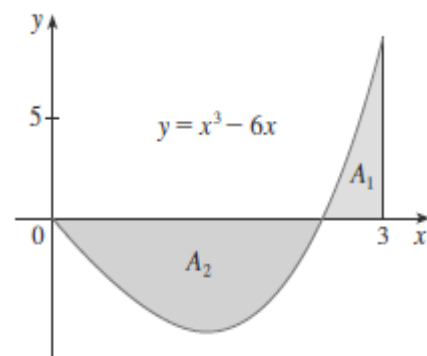
Notice that f is not a positive function and so the Riemann sum does not represent a sum of areas of rectangles. But it does represent the sum of the areas of the rectangles (above the x -axis) minus the sum of the areas of the rectangles (below the x -axis)

(b) With n subintervals we have

$$\Delta x = \frac{b - a}{n} = \frac{3}{n}$$

Thus $x_0 = 0$, $x_1 = 3/n$, $x_2 = 6/n$, $x_3 = 9/n$, and, in general, $x_i = 3i/n$. Since we are using right endpoints, we can use Equation 3:

$$\begin{aligned}\int_0^3 (x^3 - 6x) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{3i}{n}\right) \frac{3}{n} \\&= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{i=1}^n \left[\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right) \right] \\&= \lim_{n \rightarrow \infty} \frac{3}{n} \sum_{i=1}^n \left[\frac{27}{n^3} i^3 - \frac{18}{n} i \right] \\&= \lim_{n \rightarrow \infty} \left[\frac{81}{n^4} \sum_{i=1}^n i^3 - \frac{54}{n^2} \sum_{i=1}^n i \right] \\&= \lim_{n \rightarrow \infty} \left\{ \frac{81}{n^4} \left[\frac{n(n+1)}{2} \right]^2 - \frac{54}{n^2} \frac{n(n+1)}{2} \right\} \\&= \lim_{n \rightarrow \infty} \left[\frac{81}{4} \left(1 + \frac{1}{n} \right)^2 - 27 \left(1 + \frac{1}{n} \right) \right] \\&= \frac{81}{4} - 27 = -\frac{27}{4} = -6.75\end{aligned}$$



$$\int_0^3 (x^3 - 6x) dx = A_1 - A_2 = -6.75$$

EXAMPLE

Set up an expression for $\int_1^3 e^x dx$ as a limit of sums.

Here we have $f(x) = e^x$, $a = 1$, $b = 3$, and

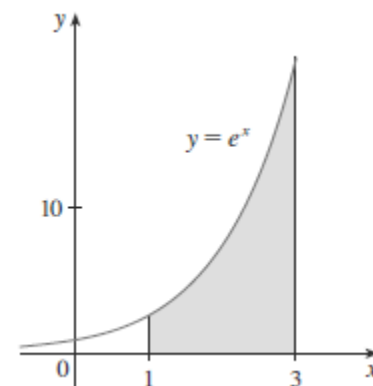
$$\Delta x = \frac{b - a}{n} = \frac{2}{n}$$

So $x_0 = 1$, $x_1 = 1 + 2/n$, $x_2 = 1 + 4/n$, $x_3 = 1 + 6/n$, and

$$x_i = 1 + \frac{2i}{n}$$

From Equation 3, we get

$$\begin{aligned}\int_1^3 e^x dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(1 + \frac{2i}{n}\right) \frac{2}{n} \\ &= \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n e^{1+2i/n}\end{aligned}$$



|||| Properties of the Definite Integral

When we defined the definite integral $\int_a^b f(x) dx$, we implicitly assumed that $a < b$. But the definition as a limit of Riemann sums makes sense even if $a > b$. Notice that if we reverse a and b , then Δx changes from $(b - a)/n$ to $(a - b)/n$. Therefore

$$\int_b^a f(x) dx = -\int_a^b f(x) dx$$

If $a = b$, then $\Delta x = 0$ and so

$$\int_a^a f(x) dx = 0$$

We now develop some basic properties of integrals that will help us to evaluate integrals in a simple manner. We assume that f and g are continuous functions.

Properties of the Integral

1. $\int_a^b c \, dx = c(b - a)$, where c is any constant
2. $\int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$
3. $\int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx$, where c is any constant
4. $\int_a^b [f(x) - g(x)] \, dx = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$

EXAMPLE Use the properties of integrals to evaluate $\int_0^1 (4 + 3x^2) \, dx$.

SOLUTION Using Properties 2 and 3 of integrals, we have

$$\int_0^1 (4 + 3x^2) \, dx = \int_0^1 4 \, dx + \int_0^1 3x^2 \, dx = \int_0^1 4 \, dx + 3 \int_0^1 x^2 \, dx$$

We know from Property 1 that

$$\int_0^1 4 \, dx = 4(1 - 0) = 4$$

and we found

that $\int_0^1 x^2 \, dx = \frac{1}{3}$. So

$$\begin{aligned} \int_0^1 (4 + 3x^2) \, dx &= \int_0^1 4 \, dx + 3 \int_0^1 x^2 \, dx \\ &= 4 + 3 \cdot \frac{1}{3} = 5 \end{aligned}$$

If $a \leq b$ and $f(x) \leq g(x)$ for $a \leq x \leq b$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

The **triangle inequality** for sums extends to definite integrals. If $a \leq b$, then

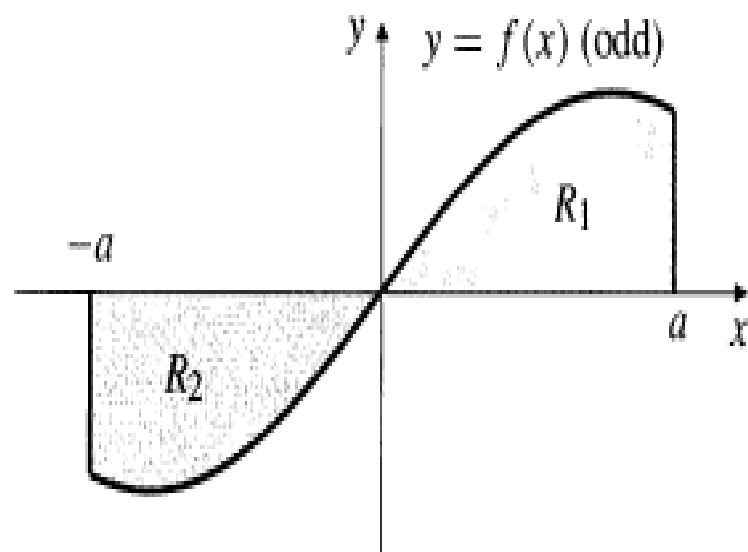
$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

The integral of an odd function over an interval symmetric about zero is zero. If f is an odd function (i.e., $f(-x) = -f(x)$), then

$$\int_{-a}^a f(x) dx = 0.$$

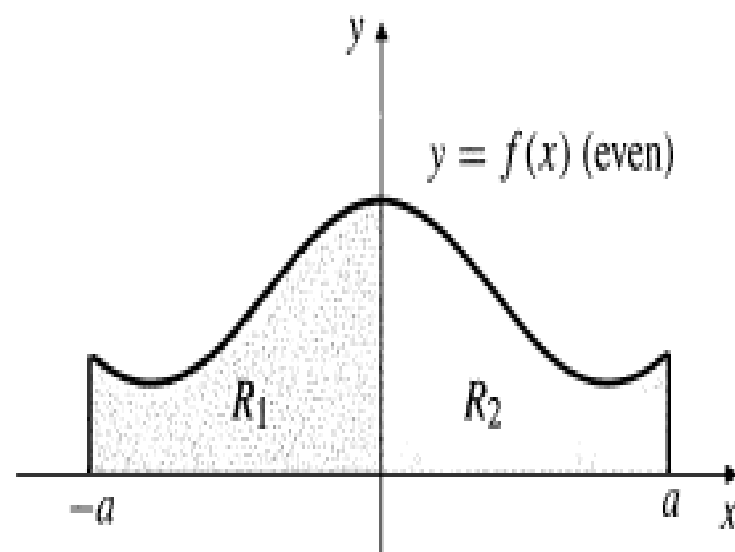
The integral of an even function over an interval symmetric about zero is twice the integral over the positive half of the interval. If f is an even function (i.e., $f(-x) = f(x)$), then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$



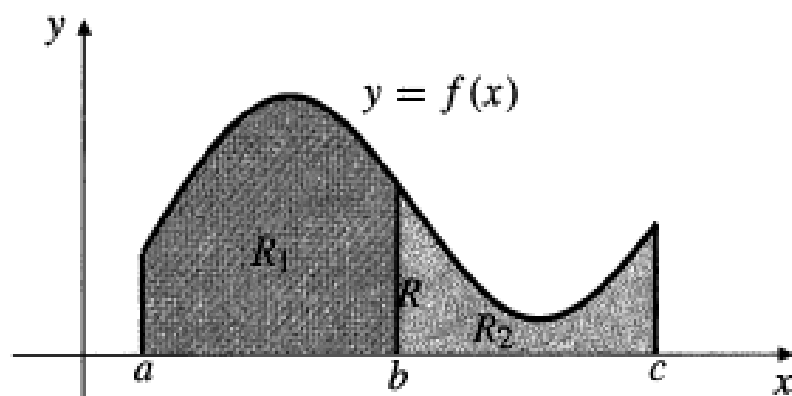
$$\text{area } R_1 - \text{area } R_2 = 0$$

$$\int_{-a}^a f(x) dx = 0$$



$$\text{area } R_1 + \text{area } R_2 = 2 \text{ area } R_2$$

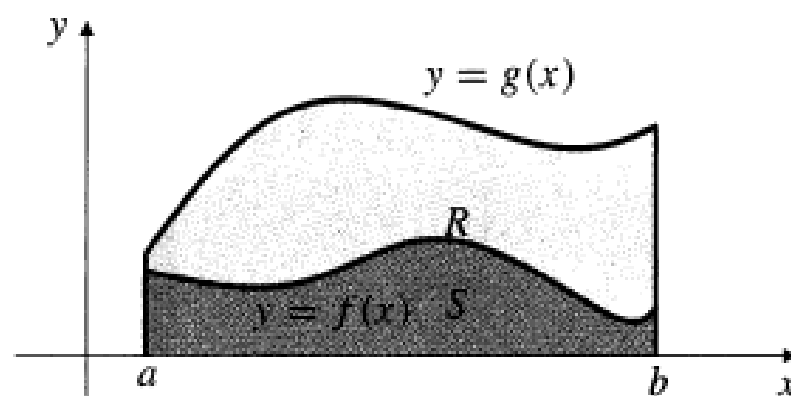
$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$



area R_1 + area R_2 = area R

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

(a)



area $S \leq$ area R

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

(b)

generalization of the triangle inequality for numbers:

$$|x + y| \leq |x| + |y|, \quad \text{or more generally,} \quad \left| \sum_{i=1}^n x_i \right| \leq \sum_{i=1}^n |x_i|.$$

$$5. \quad \int_a^c f(x) \, dx + \int_c^b f(x) \, dx = \int_a^b f(x) \, dx$$

EXAMPLE If it is known that $\int_0^{10} f(x) \, dx = 17$ and $\int_0^8 f(x) \, dx = 12$, find $\int_8^{10} f(x) \, dx$.

SOLUTION By Property 5, we have

$$\int_0^8 f(x) \, dx + \int_8^{10} f(x) \, dx = \int_0^{10} f(x) \, dx$$

$$\text{so} \quad \int_8^{10} f(x) \, dx = \int_0^{10} f(x) \, dx - \int_0^8 f(x) \, dx = 17 - 12 = 5$$

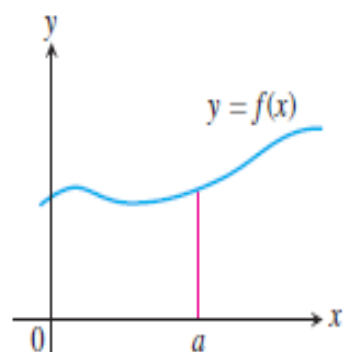
Comparison Properties of the Integral

6. If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x) \, dx \geq 0$.

7. If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) \, dx \geq \int_a^b g(x) \, dx$.

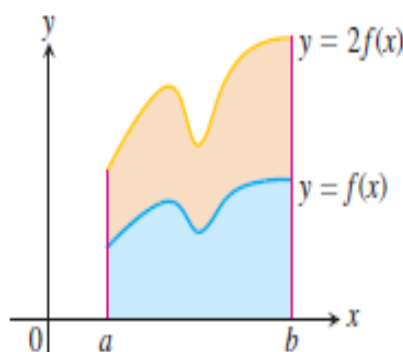
8. If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b - a) \leq \int_a^b f(x) \, dx \leq M(b - a)$$



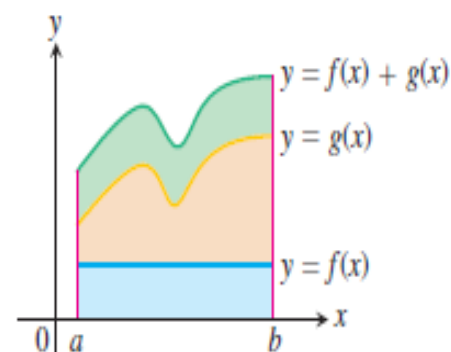
(a) Zero Width Interval:

$$\int_a^a f(x) dx = 0$$



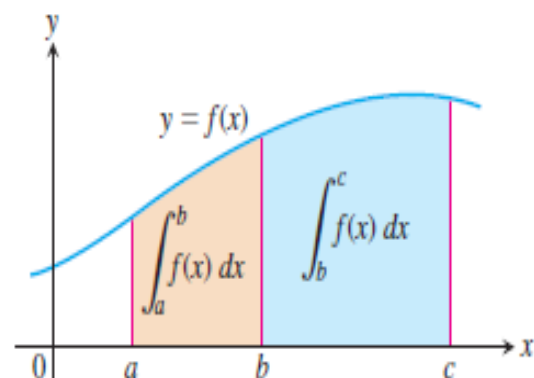
(b) Constant Multiple: ($k = 2$)

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx$$



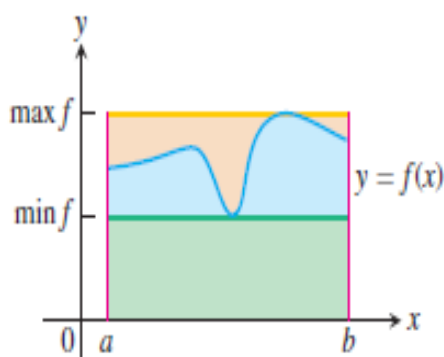
(c) Sum: (areas add)

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$



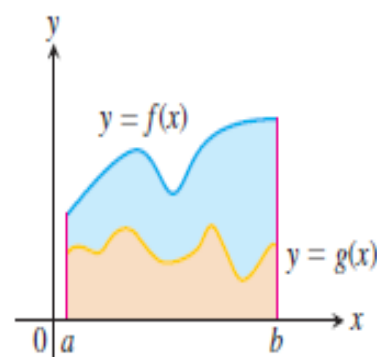
(d) Additivity for definite integrals:

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$



(e) Max-Min Inequality:

$$\begin{aligned} \min f \cdot (b - a) &\leq \int_a^b f(x) dx \\ &\leq \max f \cdot (b - a) \end{aligned}$$



(f) Domination:

$$\begin{aligned} f(x) &\geq g(x) \text{ on } [a, b] \\ \Rightarrow \int_a^b f(x) dx &\geq \int_a^b g(x) dx \end{aligned}$$

$$\int_{-1}^1 f(x) \, dx = 5, \quad \int_1^4 f(x) \, dx = -2, \quad \text{and} \quad \int_{-1}^1 h(x) \, dx = 7.$$

Then

$$1. \quad \int_4^1 f(x) \, dx = -\int_1^4 f(x) \, dx = -(-2) = 2$$

$$\begin{aligned} 2. \quad \int_{-1}^1 [2f(x) + 3h(x)] \, dx &= 2 \int_{-1}^1 f(x) \, dx + 3 \int_{-1}^1 h(x) \, dx \\ &= 2(5) + 3(7) = 31 \end{aligned}$$

$$3. \quad \int_{-1}^4 f(x) \, dx = \int_{-1}^1 f(x) \, dx + \int_1^4 f(x) \, dx = 5 + (-2) = 3$$

- (a) Express $\int_1^2 (x^2 - 1) dx$ as a Riemann sum with n sample points. (Take the sample points to be the *right* end points.)
- (b) Evaluate the sum in the limit as $n \rightarrow \infty$.

(Some useful identities:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2.)$$

- (a) To set up the n -th Riemann sum, we divide the interval $[1, 2]$ into subintervals of width $\Delta x = \frac{2-1}{n} = \frac{1}{n}$. The end points of these intervals are given by $x_i = 1 + i \cdot \Delta x = 1 + \frac{i}{n}$. Hence, the Riemann sum corresponding to the choice of sample points $x_i^* = x_i$ is

$$\begin{aligned} R_n &= \sum_{i=1}^n f(x_i) \cdot \Delta x = \sum_{i=1}^n (x_i^2 - 1) \cdot \Delta x = \sum_{i=1}^n \left[\left(1 + \frac{i}{n} \right)^2 - 1 \right] \cdot \frac{1}{n} \\ &= \sum_{i=1}^n \left[1 + \frac{2i}{n} + \frac{i^2}{n^2} - 1 \right] \cdot \frac{1}{n} = \frac{1}{n^2} \cdot \sum_{i=1}^n \left[2i + \frac{i^2}{n} \right]. \end{aligned}$$

(b) To evaluate $\lim_{n \rightarrow \infty} R_n$, use the given identities to compute

$$\begin{aligned}\lim_{n \rightarrow \infty} R_n &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \left[2 \sum_{i=1}^n i + \frac{1}{n} \sum_{i=1}^n i^2 \right] = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \left[2 \frac{n(n+1)}{2} + \frac{1}{n} \frac{n(n+1)(2n+1)}{6} \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \left[n^2 + n + \frac{2n^2 + 3n + 1}{6} \right] = \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n} + \frac{2}{6} + \frac{3}{6n} + \frac{1}{6n^2} \right] = 1 + \frac{1}{3} = \frac{4}{3}.\end{aligned}$$

Suppose $f(x) = x^2$. Let R be the region in the xy -plane bounded by the curve $y = f(x)$ and the lines $y = 0$, $x = 2$, and $x = 3$. Find the area of R by evaluating the limit of a Riemann sum that uses the *Right Endpoint Rule*; show all reasoning.

Since $f(x)$ is continuous on the interval $[2, 3]$, it is Riemann integrable. Thus we can choose the following sequence of partitions of $[2, 3]$. For each n , we let

$$I_i^{(n)} = [x_{i-1}^{(n)}, x_i^{(n)}] := \left[2 + \frac{i-1}{n}, 2 + \frac{i}{n}\right]$$

$i = 1, 2, \dots, n$. Notice that each subdivision has length $|I_i^{(n)}| = 1/n$, which goes to 0 as n goes to infinity. Hence the Riemann sum evaluated over such partitions will converge to the actual Riemann integral, which we take to be the definition of the area of R . The right endpoint of each subinterval $I_i^{(n)}$ is clearly $x_i^{(n)} = 2 + \frac{i}{n}$. By definition of Riemann integral, we have

$$\begin{aligned} R &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^{(n)}) |I_i^{(n)}| \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(2 + \frac{i}{n}\right)^2 \frac{1}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(4 + \frac{4i}{n} + \frac{i^2}{n^2}\right) \frac{1}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4}{n} + \sum_{i=1}^n \frac{4i}{n^2} + \sum_{i=1}^n \frac{i^2}{n^3} \end{aligned}$$

By linearity of limit, we evaluate the limit of the three sums separately. Recall the formula for $\sum_{i=1}^n i$ and $\sum_{i=1}^n i^2$, we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4}{n} = \lim_{n \rightarrow \infty} n \frac{4}{n} = 4$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4i}{n^2} = \lim_n \frac{4}{n^2} \sum_{i=1}^n i = \lim_n \frac{4}{n^2} \frac{n(n+1)}{2} = 2$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{i=1}^n i^2 = \lim_n \frac{1}{n^3} \frac{n(n+1)(2n+1)}{6} = 1/3.$$

So the area $R = 4 + 2 + 1/3 = 19/3$.

Example Compute the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x$ for the function $f(x) = \frac{1}{x}$ on $[1, 6]$ with a regular partition into $n = 5$ subintervals, and with $x_i^* = x_i$.

Solution: Note that $a = 1$, $b = 6$ and $n = 5$. Compute the following

$$\Delta x = \frac{b-a}{n} = \frac{6-1}{5} = 1.$$

$$x_i = a + i\Delta x = 1 + i, \text{ for each } i.$$

$$f(x_i^*) = f(x_i) = \frac{1}{1+i}, \text{ for each } i.$$

$$\sum_{i=1}^5 f(x_i^*)\Delta x = \sum_{i=1}^5 \frac{1}{i+1} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}.$$

Example Compute the integral $\int_0^4 x^3 dx$ by computing Riemann sums for a regular partition.

Solution: Note that $a = 0$, $b = 4$ and $f(x) = x^3$. Use a regular partition for each positive integer n . Note that when $n \rightarrow \infty$, $|P| \rightarrow 0$. Compute the following

$$\begin{aligned}\Delta x &= \frac{b-a}{n} = \frac{4-0}{n} = \frac{4}{n}, \\ x_i &= a + i\Delta x = \frac{4i}{n}, \text{ for each } i. \\ f(x_i^*) &= f(x_i) = \left(\frac{4i}{n}\right)^3 = 64\frac{i^3}{n^3}, \text{ for each } i.\end{aligned}$$

Therefore, the corresponding Riemann sum becomes (note that $\frac{1}{n^4}$ is viewed as constant with respect to the index i , and so it can be moved out of the summation sign. The last step follows from summation formulas)

$$\sum_{i=1}^n f(x_i^*)\Delta x = \sum_{i=1}^n 64\frac{i^3}{n^3}\frac{4}{n} = \frac{256}{n^4} \sum_{i=1}^n i^3 = \frac{256}{n^4} \frac{n^2(n+1)^2}{4}.$$

Thus the answer is

$$\int_0^4 x^3 dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n 64\frac{i^3}{n^3}\frac{4}{n} = \lim_{n \rightarrow \infty} \frac{256}{n^4} \frac{n^2(n+1)^2}{4} = 64.$$